

Sang Hyuk Hong

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EDUCATION

Korea University

B.S. Physics

GPA: 4.09 / 4.50 (Major GPA: 4.19 / 4.50)

March 2019 – February 2023

Seoul, Republic of Korea

EMPLOYMENT

Seoul National University

Research Assistant

July 2026 – Present

Seoul, Republic of Korea

- Full-time research assistant contributing to the construction of an Rb–Yb mixed-species neutral-atom tweezer array platform for quantum information science.

Agency for Defense Development

Research Officer for National Defense

June 2023 – May 2026

Daejeon, Republic of Korea

- Three-year military service as a commissioned officer in the Republic of Korea Air Force, conducted at Korea's principal national defense R&D institute; contributed to development of optically pumped magnetometers and nuclear magnetic resonance gyroscopes. Discharged as First Lieutenant.

PUBLICATIONS

- [1] **Laser mode-hopping assisted all-optical single beam pulsed atomic magnetometer**
Ji Hoon Yoon, **Sang Hyuk Hong**, Taek Jeong, Sin Hyuk Yim, Kyu Min Shim, and Sangkyung Lee
arXiv:2404.01874 (2024). [\[Link\]](#).
- [2] **Fabrication of an Rb–Xe gas cell with optical components attached by optical contact bonding for an atom spin gyroscope**
Sang Hyuk Hong, Sin Hyuk Yim, Taek Jeong, Jeong Bin Nam, and Sangkyung Lee
Applied Optics (2025). [\[Link\]](#).
- [3] **Aging test of atomic vapor cell with Al₂O₃ wall coating on cubic glass**
Hyeonjae Kim, Taek Jeong, **Sang Hyuk Hong**, Jeong Bin Nam, Sangkyung Lee, Younghoon Lim, and Sin Hyuk Yim
Applied Optics (2025). [\[Link\]](#).
- [4] **Novel sealing of glass-blown atomic gas cells using a resistive heating wire**
Sang Hyuk Hong, Younghoon Lim, Sin Hyuk Yim, Sangkyung Lee, Hyeonjae Kim, Taek Jeong, and Jeong Bin Nam
Manuscript under review (2026).
- [5] **Development and characterization of atom spin gyroscopes**
Sang Hyuk Hong, Sangkyung Lee, Sin Hyuk Yim, Sang Rae Yeo, Sangmin Lee, Hoon Yu, Taek Jeong, Jeong Bin Nam, Dongkyu Kim, and Younghoon Lim
In preparation (2026).
- [6] **AC magnetic field gradient-induced transverse relaxation for diffusive spin-1/2 particles in a cubic cell**
Sang Hyuk Hong and Sangkyung Lee
In preparation (2026).

HONORS & SCHOLARSHIPS

National Scholarship for Natural Science and Engineering

Korea Student Aid Foundation (KOSAF)

Spring 2021 – Fall 2022

Republic of Korea

- Selected as one of 23 Research Officer for National Defense scholars nationwide.

- Full tuition and stipend of $\sim$ \\$20,000.

Freshman Honors Scholarship

Department of Physics, Korea University

Fall 2019

Seoul, Republic of Korea

- Awarded to the top 2 incoming physics majors; one-semester stipend.

RESEARCH EXPERIENCE

Quantum Gas Lab, Seoul National University

Research Assistant

July 2026 – Present

Seoul, Republic of Korea

- Contributed to the construction of an experimental platform for mixed-species (alkali + alkaline-earth-like) neutral-atom tweezer arrays using Rb and Yb, including optical system assembly, ultra-high vacuum integration, and frequency-stabilized laser sources.
- Targeted physics includes coherent tunneling in excited states at reduced tweezer spacings, and impurity physics of individual atoms immersed in a species-selective optical lattice, where one species acts as a controllable environment for the other.

Agency for Defense Development

Research Officer for National Defense

June 2023 – May 2026

Daejeon, Republic of Korea

Optically Pumped Magnetometer (OPM)

- Contributed to development of a single-beam pulsed-mode OPM scheme in which DBR laser mode-hopping is repurposed as an internal $\sim$ 20 GHz frequency jump to drive discontinuous synchronous optical pumping without any external modulator [1].
- Analyzed noise characteristics and heading errors of the mode-hop assisted OPM against conventional continuous-mode schemes.
- Developed a vector magnetometry technique exploiting the AC Stark shift in the mode-hopping assisted scalar OPM (Korean patent filed).
- Extended the platform toward magnetic-anomaly detection, magnetic navigation, and multi-channel magnetic imaging (manuscripts in preparation).

Nuclear Magnetic Resonance Gyroscope (NMRG)

- Fabricated a monolithic Rb–Xe vapor cell in which prism mirrors and a reflection mirror are bonded directly to a cubic cell by optical contact bonding, enabling a compact, glue-free optical assembly for low-SWaP NMRGs [2].
- Developed a precision sealing method using a resistive heating wire for glass-blown atomic gas cells that suppresses geometric asymmetry—one of the major sources for quadrupole-interaction-induced systematics in alkali–noble-gas systems (Korean patent filed; related manuscript under review [3]).
- Performed numerical studies of coil-geometry magnetic uniformity and heater-induced thermal distribution (COMSOL) informing low-SWaP NMRG design; results incorporated into an ongoing device-level performance manuscript [4].
- Developed an analytical theory, and experimentally verified, AC magnetic-field-gradient-induced transverse relaxation of diffusive spin-1/2 particles in a cubic cell, providing a quantitative framework for ^{129}Xe dephasing under inhomogeneously polarized Rb ensembles (manuscript in preparation) [5].

Korea University

Summer Research Intern

June 2022 – August 2022

Seoul, Republic of Korea

- Joined a summer undergraduate research internship under a Samsung Science & Technology Foundation-supported project on continuum dark matter, gaining introductory exposure to theoretical models beyond the conventional WIMP framework and their phenomenological implications.

PATENTS

- **Heated-wire sealing method for atomic vapor cells**, Korean patent application filed, 2025.

- Vector magnetic-field measurement method using AC Stark effect in a scalar optically pumped magnetometer, Korean patent application filed, 2026.

PRESENTATIONS

2023 Korean Physical Society Fall Meeting

October 2023

Poster Presentation

Changwon, Republic of Korea

- Optimization of Saddle Coil for Uniform Magnetic Field in Atomic Magnetometer Experiment.

2024 Spring Conference of the Korean Sensors Society

April 2024

Poster Presentation

Daejeon, Republic of Korea

- Nullifying ^{87}Rb Polarization-Induced Magnetic-Field Gradients via Modulation of Pump-Beam Polarization.

55th APS DAMOP Meeting

June 2024

Poster Presentation

Fort Worth, Texas, USA

- Analysis of Heading Errors in Pulsed-Mode and Continuous-Mode All-Optical Atomic Magnetometers.

16th CLEO-PR

August 2024

Poster Presentation

Incheon, Republic of Korea

- Atom Spin Gyroscope Using an Rb–Xe Gas Cell.

2024 Korean Physical Society Fall Meeting

October 2024

Poster Presentation

Yeosu, Republic of Korea

- T_1 Estimation of ^{129}Xe from a Single Free-Induction-Decay Measurement Based on Pump-Beam Modulation.

2025 KIMST Annual Conference

June 2025

Poster Presentation

Jeju Island, Republic of Korea

- High-Symmetry Glass-Blown Atomic Gas Cell Fabricated via a Heated-Wire Precision Sealing Method.

2025 Korean Physical Society Fall Meeting

October 2025

Oral Presentation

Gwangju, Republic of Korea

- AC Magnetic-Field-Gradient-Induced Transverse Relaxation for Diffusive Spin-1/2 Particles in a Cubic Cell.

2025 N³etFraST Workshop

November 2025

Invited Oral Presentation

Seoul, Republic of Korea

- AC Magnetic-Field-Gradient-Induced Transverse Relaxation for Diffusive Spin-1/2 Particles in a Cubic Cell.

2026 Optical Society of Korea Winter Meeting

February 2026

Poster Presentation

Busan, Republic of Korea

- Development and Characterization of Atom Spin Gyroscopes.

TECHNICAL SKILLS

Programming Languages

Python, Wolfram Mathematica.

Scientific Software

COMSOL Multiphysics, LabVIEW, Origin, SolidWorks.